



VRIJE
UNIVERSITEIT
BRUSSEL



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degree of de Ingenieurswetenschappen: Computerwetenschappen

PROGRAMMING ROBOTS WITH HIGHER-ORDER REACTIVE PROGRAMMING

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2025-2026

Promotor: Prof. Dr. Wolfgang De Meuter
Advisor: Bjarno Oeyen

Sciences and Bio-engineering Sciences



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Wetenschappen en Bio-ingenieurswetenschappen

Abstract

Your abstract would go here.

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Chapter 1

Introduction

In today's world, avoiding electronics or their applications is unthinkable.

1.1 Contributions

1.2 Overview of this Dissertation

1.3 Approach

Chapter 2

State of the Art

2.1 Reactive Programming

2.2 RP languages targeting MCUs

2.3 Problem Statement

Chapter 3

μ -Remus

3.1 Haai & Remus

3.2 ROS Compatability issues

3.3 μ -Remus: A C implementation

Chapter 4

Programming Robots the Reactive Way

4.1 ROS communication protocols

4.2 Linking ROS and μ -ROS

Chapter 5

Tierless Programming

5.1 Declaring Haai's locality

5.2 Extending Remus

Evaluation

Conclusion & Future Work